

Bao Vo

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EDUCATION

University of Michigan, Ann Arbor, MI

Apr 2026

Master of Science in Electrical and Computer Engineering - Focus in Integrated Circuits & VLSI

GPA: 3.68/4.00

- **Honors & Awards:** Fully Funded ECE Doctoral Fellowship, Graduate Certificate in Innovation & Entrepreneurship
- **Relevant Advanced Coursework:** VLSI Design I & II, Integrated Analog/Digital Interface Circuits, Analog Integrated Circuits, VLSI for Wireless Communication and Machine Learning, Control Systems Analysis and Design

University of Michigan, Ann Arbor, MI

Dec 2023

Bachelor of Science in Materials Science and Engineering and Electrical Engineering Minor

GPA: 3.89/4.00

- **Honors & Awards:** Summa Cum Laude, Departmental Scholarships, Pan-Asia Alumni Scholarship, Dean's List

WORK EXPERIENCE

Project Manager and Veteran Life Science Consultant

Oct 2024 – May 2026

miLEAD Consulting | University of Michigan

- Developed market entry and pivot strategy for 3 pre-clinical biotechnology startups through primary research
- Utilized Pitchbook and market databases to present strategic recommendations directly to the leadership team on market positioning, pricing strategy, and target segments, driving pivots toward less saturated, higher-fit markets

Graduate Student Instructor

Aug 2025 – May 2026

Department of Electrical and Computer Engineering | University of Michigan

- Lectured, hosted office hours, and co-developed laboratory circuit/amplifier materials for 400+ students in EECS 215

Graduate Student Researcher

Jul 2024 – Nov 2025

Wireless Integrated Circuits and Systems Group | University of Michigan

- Designed ultra-low-power circuits for implantable medical devices, focusing on a totally implantable auditory prosthesis featuring ultra-low-power subthreshold design with flip chip packaging
- Developed circuit blocks for the MEMS/ASIC interface, transimpedance amplifiers, and active RC filters for a low-impedance, low-noise frontend system for an audio range application in TSMC 65nm
- Validated circuit stability and noise performance across varied PVT corners using Cadence Virtuoso and SPICE

DFT/STA Engineer

Jan 2024 – May 2024

Faraday Technology Corporation | Ho Chi Minh City, Vietnam

- Taped out an ultra-low-power chip in UMC 22nm for an ASIC shuttle project, driving chiptop DFT/STA flows to successful completion within an aggressive 1.5-month timeline while earning a top 10% performance evaluation
- Defined DFT specifications and strategy to meet customer test coverage targets and implemented the architecture from RTL to GDSII, including scan chain insertion, MBIST integration, simulation, and ATE patterns delivery
- Built and maintained automation flows (Tcl/Perl/Csh) for DFT and STA spanning pre-synthesis checks, ATPG, and post-layout verification, reducing manual effort and improving flow reliability
- Utilized PrimeTime to debug STA timing violations and provided recommendations to the physical design team
- Mentored senior undergraduates in a 2-month internship program, most of whom later converted to full-time roles

SELECTED PAST PROJECTS

Designing an NTT Accelerator for Post-Quantum Cryptography | University of Michigan

EECS 627

Jan 2026 – Apr 2026

- Designed a reconfigurable radix-2 NTT accelerator for polynomial sizes $n=256/1024$ using a Pease architecture with 16 parallel butterfly processing elements, resolving bandwidth bottlenecks via a 16-bank dual-ported SRAM array
- Implemented AXI-4 DMA/FIFO integration supporting out-of-order burst transactions (~3.6 GB/s throughput)
- Completed synthesis and APR of full-chip design (10 mm × 8.4 mm), achieving 100 MHz and ~298 kNTT/s

Designing a Quadrature Phase Mixer-First Radio Frequency Front-end | University of Michigan

Advisor: Professor David Wentzloff

May 2023 – Dec 2023

- Modeled and developed a Windmill frequency divider for low-power, low-phase-noise quadrature clock generation
- Designed a mixer-first quadrature phase front-end RF receiver using TSMC 65nm technology and Virtuoso ADE on Cadence, optimizing micro-level power consumption with good phase noise performance

SKILL SET

- **EDA/CAD Tools:** Cadence Virtuoso, Layout, SPICE, Design Compiler, Innovus, PrimeTime, NCVerilog, Verdi
- **Software Skills:** Tcl, Perl, Csh, MATLAB, Python, C++, C, Verilog, SystemVerilog, Linux, Git
- **Lab & Engineering:** Oscilloscope, Power Supply, Soldering, Multimeter, PPMS, SEM, Thin-Film Evaporators